

Food-intake detection from AIM-2 sensor signals: literature review and model description

Caelan Dunlea
University of Alabama

1 October 2026

1 Literature review

Objective measurement of eating behaviour matters for obesity, eating disorders and diabetes research, and self-report is known to be inaccurate and cannot capture in-meal behaviour such as eating rate or pauses (1). Wearable sensors that detect chewing, bites or hand gestures have therefore become a central approach, ranging from classical machine learning on hand-crafted features to end-to-end deep learning, with the aims of improving accuracy, reducing false detections and generalising to people outside the training set (2). Fontana et al. (3) introduced the Automatic Ingestion Monitor (AIM), combining jaw-motion, hand-gesture and accelerometer sensors, and reported 89.8% food-intake detection accuracy over 24 hours of free living. Its successor, AIM-2 (4), moved the sensors onto eyeglasses: a temporalis-muscle sensor and a 3-axis accelerometer at 128 Hz. It extracted 38 hand-crafted features per signal, selected nine by minimum-redundancy maximum-relevance, and classified 10 s epochs with a support vector machine, reaching F1 0.818 under leave-one-subject-out validation on 30 participants, as summarised in Table 1. These feature-engineered models work, but their performance is bounded by the features chosen by hand, and they are sensitive to noise and to variation between people.

Table 1: State-of-the-art methods for food-intake and chewing detection, and the two ECG studies whose methods transfer to this problem.

References	Techniques	Goals	Findings
Doulah et al. (4)	AIM-2; hand-crafted features, mRMR selection, SVM	Food-intake detection with passive food-image capture	F1 0.818 (LOSO, 30 participants); camera triggered only during eating
Diou et al. (1)	Review: CNN on in-ear audio windows for chewing, smartwatch IMU for bites	Intake monitoring in free living	CNN F1 0.908 against 0.748 for an SVM; self-supervised learning comparable to supervised

References	Techniques	Goals	Findings
Kyritsis et al. (5)	CNN + LSTM, end to end on raw wrist IMU	Bite and meal detection in the wild	Bite F1 0.923 (LOSO)
Ghosh and Sazonov (6)	Five deep networks (MLP, time-CNN, FCN, ResNet, Inception) on AIM-2 optical and accelerometer time series	Compare deep architectures for food-intake detection	ResNet best, balanced accuracy 93.47%
Ghosh et al. (7)	Wavelet scalogram of accelerometer, fused with food-image recognition	Reduce false positives in free living	Fused F1 80.77%; fusion cuts false positives
Stankoski et al. (8)	Selection of hard negative examples for training	Imbalanced data with imperfect labels	Person-independent F1 0.82
Wang et al. (9)	Temporal CNN + multi-head self-attention	Eating speed from wrist IMU, towards free living	61 participants, 513 h; eating-speed MAPE 0.110
Vedovelli et al. (10)	Transformer encoder and other models on smartwatch sensors	Eating-episode recognition	Transformer: sensitivity 0.691, specificity 0.504 (LOSO)
Hannun et al. (11)	34-layer 1D CNN on raw single-lead ECG	End-to-end arrhythmia classification	F1 0.837, above the average cardiologist (0.780)
Ikram et al. (12)	Transformer on MIT-BIH beats	Five-class arrhythmia classification	Accuracy 97.1%; split not at patient level

1.1 From hand-crafted features to learned representations

To address the limits of hand-crafted features, several groups trained networks directly on the signal. In this literature, a raw input denotes the absence of hand-engineered features rather than of preprocessing: z-normalisation is the only preprocessing in the convolutional time-series baselines of Wang et al. (13), and the end-to-end bite detector reviewed by Diou et al. (1) receives preprocessed inertial signals. Papapanagiotou et al. (14) detected chewing with a 1D convolutional network applied to in-ear audio windows; across window lengths of 1 to 5 s, the 5 s network reached F1 0.908, against 0.748 for a support vector machine on hand-crafted features, although such models “require significantly larger volumes of labeled data” (1). Similarly, Kyritsis et al. (5) trained a CNN feature extractor and an LSTM jointly on raw wrist inertial data, reaching a bite-detection F1 of 0.923 under leave-one-subject-out evaluation. On AIM-2 itself, Ghosh and Sazonov (6) compared five deep architectures on the optical and accelerometer time series and found the residual network best, with a balanced accuracy of 93.47%. The same pattern holds outside eating: the fully convolutional network of Wang et al. (13), three blocks of convolution, batch normalisation and ReLU, is a strong baseline across time-series benchmarks.

The closest analogue to this problem is electrocardiography. Hannun et al. (11) classified arrhythmia with a 34-layer 1D CNN applied end to end to raw single-lead ECG at 200 Hz, and exceeded the

average cardiologist (F1 0.837 against 0.780). ECG and the AIM-2 signals share a sampling rate of the same order, a quasi-periodic structure (heart beats at about 1 Hz, chews at 0.94–2.17 Hz (15)) and the need to recognise short events inside longer recordings. This motivates the first and most important design choice in this work: the model reads the normalised time series, not a spectrum. None of the reviewed studies compares the two representations on the same network, so that comparison is made directly in this work. Ghosh et al. (7) took the opposite route, converting accelerometer signals to a wavelet scalogram image. A magnitude spectrum says which rhythms are present in a window but discards when they occur, which is the information that separates a bite from background motion; a scalogram keeps a time axis, but at a resolution fixed by the transform rather than learned from the data.

1.2 Attention and Transformers

Attention mechanisms let a network weight the parts of its input that carry evidence. Squeeze-and-excitation (16) gates feature channels with a small learned function of their global average, at negligible cost. Attention-based pooling (17) treats a window as a bag of instances and learns which instances determine its label; for eating, the instances are time steps and a single bite may be the deciding one. Transformers extend this to attention between all time steps. Ikram et al. (12) applied a Transformer to five-class arrhythmia classification on MIT-BIH, reporting 97.1% accuracy. The reported 80/20 split is of preprocessed samples rather than of patients, so the figure measures performance on new beats from known patients, not on new patients. In eating detection, Wang et al. (9) combined a temporal CNN with multi-head self-attention to measure eating speed from wrist IMU data on 61 participants, while Vedovelli et al. (10) found that a Transformer encoder on smartwatch data under leave-one-subject-out gave the highest sensitivity (0.691) but a specificity of only 0.504, which is the false-positive problem in its clearest form.

Despite their promise, Transformers carry many more parameters than the attention layers above, and whether they help at a given data size is an empirical question rather than a given. This work therefore uses the two light attention mechanisms by default and treats a Transformer over the convolutional features as a separate experiment, on the same folds.

1.3 Supervision, augmentation and negative data

Small annotated datasets call for more supervision per example and for more examples per recording. Multi-task learning (18) improves generalisation by training a shared representation on related tasks; here, counting chews is the related task, since the ground truth already contains chew annotations. This auxiliary task is specific to the present work. Augmentation of wearable sensor data raised accuracy from 77.54% to 86.88% in Parkinson’s disease monitoring (19); overlapping windows are the simplest form of it. Window length itself trades resolution for context, and has a measurable effect on recognition (20). For chewing, which “typically is in the range from 1 to 3 Hz”, a 5 s window was chosen so that “even for the lowest frequency of 1 Hz, a window of 5 s contains at least 5 periods” (1); the 8 s windows used here contain at least eight.

False positives are the persistent weakness of eating detectors: many non-eating activities produce jaw and head motion. Walking in particular has a rhythm similar to chewing, 1 to 3 Hz, and causes chewing classifiers “to yield high false-positive detections”, which accelerometer data can be used to remove (1). The same work notes that a detector can be tuned “towards higher recall or precision, or a balanced mode, based on the needs of each use-case”. Stankoski et al. (8) reduced them by deliberately selecting hard negative examples for training, and Ghosh et al. (7) by fusing the sensor

classifier with food-image recognition. Finally, evaluation practice varies widely across the field (2). Saeb et al. (21) showed that record-wise cross-validation, where windows from one person appear in both training and test sets, can massively overestimate accuracy, and that subject-wise validation is the one that matches the use case. The leave-one-subject-out protocol used by Doulah et al. (4) and Diou et al. (1) is adopted here for the same reason.

1.4 Contribution of this work

- A compact convolutional model with channel attention, temporal attention pooling and an auxiliary chew-count head, reading the normalised AIM-2 optical and accelerometer signals directly from the ETS database: 61 annotated sessions from 43 participants, evaluated at the participant level.
- A direct comparison of time-series and spectral input on the same network, which none of the reviewed studies reports. The time series reaches F1 0.858 against 0.796 and reduces false alarms from 81 to 32 per hour of non-eating.
- A controlled ablation of each design decision, one at a time on identical participant-level folds, separating the components that carry the performance from those that do not.
- Negative results reported with their mechanism: temporal median smoothing and context windows without a target marker both lower F1, because meals at 8 s resolution are punctuated rather than contiguous.
- A measured check of sensor-annotation alignment on every session, replacing an assumed correction with a verified one.

Despite the progress in deep learning for intake monitoring, several challenges persist: limited annotated data, variation between participants in sensor fit and chewing style, false positives from non-eating jaw and head motion, and inconsistent evaluation that makes published results hard to compare. Addressing these requires models that generalise to people they have not seen, evaluated in the way they would be used.

2 Model description

The model is a compact one-dimensional convolutional network that reads the normalised AIM-2 sensor signals and outputs the probability that an 8 s window contains eating (Figure 1). It has four parts. A convolutional feature extractor learns its own filters from the time series instead of receiving hand-crafted features or a spectrum. A squeeze-and-excitation gate reweights the learned feature channels in each window. An attention pooling layer weights the time steps that carry the evidence. Finally, the output has two heads: one classifies eating, and one counts chews and is used only during training. The network has 47,282 parameters at inference (Table 2). Table 3 lists every design decision with its reason and the literature it rests on.

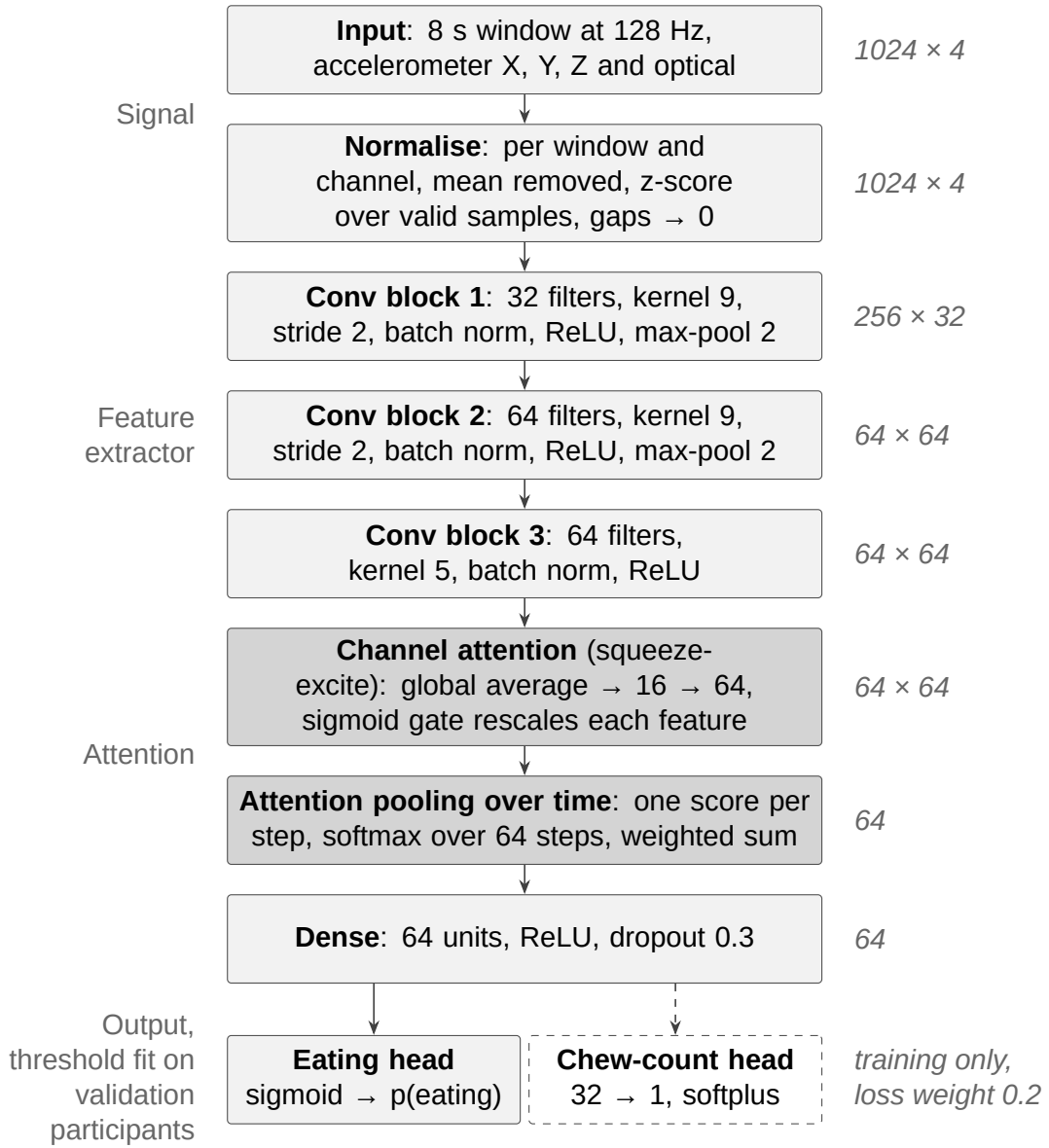


Figure 1: Model architecture. Shaded blocks are the two attention mechanisms. The dashed chew-count head is trained jointly and discarded at inference. Output shapes are time steps $\times$ features.

Table 2: Layer dimensions. An 8 s window at 128 Hz is 1024 samples; each of the 64 output time steps covers about 0.9 s of signal at 125 ms spacing.

Layer	Configuration	Output	Parameters
Input	Acc X, Y, Z, Optical	1024 × 4	–
Conv block 1	32 × k9, stride 2, BN, max-pool 2	256 × 32	1,312
Conv block 2	64 × k9, stride 2, BN, max-pool 2	64 × 64	18,752
Conv block 3	64 × k5, BN	64 × 64	20,800
Channel attention	64 → 16 → 64, sigmoid	64 × 64	2,128
Attention pooling	1 × 1 conv, softmax over time	64	65
Dense	64, ReLU, dropout 0.3	64	4,160
Eating head	1, sigmoid	1	65
<i>Total at inference</i>			47,282
Chew-count head	32 → 1, softplus (training only)	1	2,113

Table 3: Complete description of the food-intake detection model. Every design decision, the reason for it, and the literature it is grounded in.

#	Component	Design	Why	Literature
<i>Data</i>				
1	Sensor	AIM-2 eyeglass sensor: 3-axis accelerometer and optical temporalis-muscle sensor, 128 Hz, stored as 8 s packets in the ETS database	Chewing moves the temporalis muscle and the head; both channels carry it, with different artefacts. The original AIM-2 used a flex sensor where this one has the optical sensor	Doulah et al. (4), Ghosh and Sazonov (6)
2	Labels	Eating = bout OR bite annotation at 10 Hz. Window is eating if more than half its annotated samples are eating. Unrecorded time (−1) is dropped, never treated as non-eating	Treating unannotated time as negative would bury the real labels and invent easy negatives	(this work)
3	Alignment	No time shift. Sensor/annotation lag measured by cross-correlating the optical chewing envelope with the labels: median +0.10 s, all within 0.2 s on the 20 sessions measured first (to be repeated on all 61)	A correction must be measured, not assumed; a residual lag of one packet would corrupt every label edge	(this work)

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#	Component	Design	Why	Literature
4	Window	8 s (1024 samples), non-overlapping at evaluation	Chewing lies at 1–3 Hz; a 5 s window was chosen for chewing detection so that it holds at least five chews, and AIM-2 used 10 s epochs. An 8 s window holds at least eight chews and matches the packet size. Lengths of 2–16 s are also evaluated	Diou et al. (1), Doulah et al. (4), Po et al. (15)
<i>Input</i>				
5	Representation	Normalised time series, 1024 samples × 4 channels; no spectrum and no hand-crafted features	In the reviewed studies, features learned from the signal outperformed hand-engineered ones. None compares time-series and spectral input on the same network; that comparison is made here (61 sessions, 5-fold): F1 0.858 against 0.796, and 32 against 81 false alarms per hour of non-eating	Diou et al. (1), Kyritsis et al. (5), Ghosh and Sazonov (6); this work
6	Normalisation	Per window, per channel: mean removed, z-scored over valid samples only; missing samples set to 0 after scaling	Optical baseline varies by participant and fit on the face; DC level is not information about chewing. A literal –1 against counts near 3000 injects a step at every gap. z-normalisation is likewise the only preprocessing in the standard convolutional time-series baselines	Wang et al. (13)
<i>Network</i>				
7	Convolutional trunk	Three 1D conv blocks: 32@k9/s2, 64@k9/s2, 64@k5, each with batch norm and ReLU, max-pool 2 after the first two. Output 64 time steps × 64 features	Learns its own filters from the signal. For chewing detection, a CNN reached F1 0.908 against 0.748 for an SVM on hand-crafted features. Each output step covers about 0.9 s of signal (one chew cycle) at 125 ms spacing	Diou et al. (1), Ghosh and Sazonov (6), Hannun et al. (11), Wang et al. (13)

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#	Component	Design	Why	Literature
8	Channel attention	Squeeze-and-excitation: global average → dense 16 (ReLU) → dense 64 (sigmoid) → rescale each feature channel	Walking at 1–3 Hz resembles chewing and produces false positives in chewing detectors; the accelerometer is what separates the two. The gate reweights optical against motion channels in every window at negligible cost	Diou et al. (1), Hu et al. (16)
9	Temporal pooling	Attention pooling: one score per time step (1×1 conv), softmax over the 64 steps, weighted sum	Global averaging dilutes a 1–2 s bite across 8 s. Attention weights the informative steps, as in multiple-instance learning	Ilse et al. (17)
10	Classifier	Dense 64 (ReLU), dropout 0.3, sigmoid output	Small head; about 47 k parameters in total at inference	–
11	Auxiliary head	Chew-count regression (dense 32 → 1, softplus), target scaled by its 99th percentile. Used in training only	Deep models require large volumes of labelled data, and one label bit per window is thin supervision. The chew-count task is specific to this work; the principle is multi-task learning	Diou et al. (1), Caruana (18)
<i>Training</i>				
12	Loss	Binary cross-entropy (weight 1.0) + chew MSE (weight 0.2); class-balanced sample weights computed per fold	Auxiliary loss is a regulariser, not the objective. Eating fraction ranges 0–86% by session, so the rare class changes with the fold	Caruana (18)
13	Augmentation	50% overlapping windows (4 s hop) at training time only	Roughly doubles the training windows at no collection cost; evaluation stays on the non-overlapping grid so no second is scored twice	Um et al. (19)
14	Optimisation	Adam, learning rate 0.0003, batch 64, up to 80 epochs, early stopping (patience 12) and learning-rate halving on validation ROC AUC	At 0.001 the validation optimum fell at the first epoch in most folds; the lower rate moves it later. Validation participants are held out from training, never random windows	–
<i>Evaluation</i>				

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#	Component	Design	Why	Literature
15	Validation	Leave-one-subject-out: 43 folds, each holding out one participant with all of their sessions. Inner validation on further held-out participants	Windows of one meal are near-duplicates; any split below participant level leaks and overstates performance. Leave-one-subject-out is the protocol of the AIM-2 and smartwatch intake studies	Diou et al. (1), Doulah et al. (4), Kyritsis et al. (5), Saeb et al. (21)
16	Threshold	Chosen on the inner validation participants, never on the test participant: at the centre of the F1 plateau by default, or as the highest recall within a false-alarm budget per hour	The F1 curve is flat near its peak, so its exact argmax on few participants is noise. A detector can be tuned towards recall or precision according to the use case; in free living, false alarms carry the higher cost	Diou et al. (1)
17	Post-processing	None. Median smoothing over 3 windows was tested and rejected (F1 -0.079)	Meals are punctuated by pauses; at 8 s the label genuinely switches, and smoothing erases short real bouts	(this work)
18	Metrics	F1 and balanced accuracy lead; accuracy, precision, recall and false alarms per hour of non-eating reported alongside, pooled over all held-out windows and per participant	Class balance differs by participant; accuracy alone rewards predicting the majority class. False alarms per hour expresses the free-living failure mode directly	Bell et al. (2)

2.1 Dataset and current performance

The dataset is 61 annotated eating sessions (breakfast, lunch, dinner and snack) from 43 participants, 2 to 55 minutes each: 24.2 hours of annotated recording, of which 9.6 hours is eating. The eating fraction ranges from 0% to 86% by session. Validation is leave-one-subject-out: 43 folds, each holding out one participant with all of their sessions.

Under leave-one-subject-out validation, pooled over all 10,891 held-out non-overlapping windows, the model reaches F1 0.856, accuracy 0.887, balanced accuracy 0.881, precision 0.860, recall 0.853 and specificity 0.909, with 41 false alarms per hour of non-eating. Per-participant F1 has mean 0.836 (standard deviation 0.117, median 0.867), and three participants fall below 0.55. The same configuration under participant-level 5-fold cross-validation gives F1 0.864 and accuracy 0.893, so the estimate is not sensitive to the validation scheme.

The training configurations tested (early-stopping criterion, learning rate, augmentation) differ by no more than 0.006 F1, while replacing the time series with a magnitude spectrum lowers F1 from 0.858 to 0.796 and raises false alarms from 32 to 81 per hour (5-fold). The input representation therefore carries the performance, and the variation between participants is larger than any difference between

training configurations.

References

- [1] Christos Diou, Konstantinos Kyritsis, Vasileios Papapanagiotou, and Ioannis Sarafis. Intake monitoring in free-living conditions: Overview and lessons we have learned. *Appetite*, 176:106096, 2022. doi: 10.1016/j.appet.2022.106096.
- [2] Brooke M. Bell, Ridwan Alam, Nabil Alshurafa, Edison Thomaz, Abu S. Mondol, Kayla de la Haye, John A. Stankovic, John Lach, and Donna Spruijt-Metz. Automatic, wearable-based, in-field eating detection approaches for public health research: a scoping review. *npj Digital Medicine*, 3: 38, 2020. doi: 10.1038/s41746-020-0246-2.
- [3] Juan M. Fontana, Muhammad Farooq, and Edward Sazonov. Automatic ingestion monitor: A novel wearable device for monitoring of ingestive behavior. *IEEE Transactions on Biomedical Engineering*, 61(6):1772–1779, 2014. doi: 10.1109/TBME.2014.2306773.
- [4] Abul Doulah, Tonmoy Ghosh, Delwar Hossain, Masudul H. Imtiaz, and Edward Sazonov. “Automatic Ingestion Monitor Version 2” – a novel wearable device for automatic food intake detection and passive capture of food images. *IEEE Journal of Biomedical and Health Informatics*, 25(2): 568–576, 2021. doi: 10.1109/JBHI.2020.2995473.
- [5] Konstantinos Kyritsis, Christos Diou, and Anastasios Delopoulos. A data driven end-to-end approach for in-the-wild monitoring of eating behavior using smartwatches. *IEEE Journal of Biomedical and Health Informatics*, 25(1):22–34, 2021.
- [6] Tonmoy Ghosh and Edward Sazonov. A comparative study of deep learning algorithms for detecting food intake. In *44th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pages 2993–2996, 2022. doi: 10.1109/EMBC48229.2022.9871278.
- [7] Tonmoy Ghosh, Yue Han, Viprav Raju, Delwar Hossain, Megan A. McCrory, Janine Higgins, Carol Boushey, Edward J. Delp, and Edward Sazonov. Integrated image and sensor-based food intake detection in free-living. *Scientific Reports*, 14:1665, 2024. doi: 10.1038/s41598-024-51687-3.
- [8] Simon Stankoski, Marko Jordan, Hristijan Gjoreski, and Mitja Luštrek. Smartwatch-based eating detection: Data selection for machine learning from imbalanced data with imperfect labels. *Sensors*, 21(5):1902, 2021. doi: 10.3390/s21051902.
- [9] Chunzhuo Wang, T. Sunil Kumar, Walter De Raedt, Guido Camps, Hans Hallez, and Bart Vanrumste. Eating speed measurement using wrist-worn IMU sensors towards free-living environments. *IEEE Journal of Biomedical and Health Informatics*, 28(10):5816–5828, 2024.
- [10] Luca Vedovelli, M. J. Bhuyan, Corrado Lanera, Ileana Baldi, and Dario Gregori. Recognition of eating episodes via commercial smartwatch sensors. *PLOS Digital Health*, 2026. doi: 10.1371/journal.pdig.0001539.
- [11] Awni Y. Hannun, Pranav Rajpurkar, Masoumeh Haghpanahi, Geoffrey H. Tison, Codie Bourn, Mintu P. Turakhia, and Andrew Y. Ng. Cardiologist-level arrhythmia detection and classification in ambulatory electrocardiograms using a deep neural network. *Nature Medicine*, 25(1):65–69, 2019. doi: 10.1038/s41591-018-0268-3.

- [12] Sunnia Ikram, Amna Ikram, Harvinder Singh, Malik Daler Ali Awan, Sajid Naveed, Isabel De la Torre Díez, Henry Fabian Gongora, and Thania Candelaria Chio Montero. Transformer-based ECG classification for early detection of cardiac arrhythmias. *Frontiers in Medicine*, 12:1600855, 2025. doi: 10.3389/fmed.2025.1600855.
- [13] Zhiguang Wang, Weizhong Yan, and Tim Oates. Time series classification from scratch with deep neural networks: A strong baseline. In *International Joint Conference on Neural Networks (IJCNN)*, pages 1578–1585, 2017. doi: 10.1109/IJCNN.2017.7966039.
- [14] Vasileios Papapanagiotou, Christos Diou, and Anastasios Delopoulos. Chewing detection from an in-ear microphone using convolutional neural networks. In *39th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pages 1258–1261, 2017. doi: 10.1109/EMBC.2017.8037060.
- [15] J. M. C. Po, J. A. Kieser, L. M. Gallo, A. J. Tésenyi, P. Herbison, and M. Farella. Time-frequency analysis of chewing activity in the natural environment. *Journal of Dental Research*, 90(10), 2011. doi: 10.1177/0022034511416669.
- [16] Jie Hu, Li Shen, and Gang Sun. Squeeze-and-excitation networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7132–7141, 2018. doi: 10.1109/CVPR.2018.00745.
- [17] Maximilian Ilse, Jakub M. Tomczak, and Max Welling. Attention-based deep multiple instance learning. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, volume 80 of *Proceedings of Machine Learning Research*, pages 2127–2136, 2018.
- [18] Rich Caruana. Multitask learning. *Machine Learning*, 28:41–75, 1997. doi: 10.1023/A:1007379606734.
- [19] Terry T. Um, Franz M. J. Pfister, Daniel Pichler, Satoshi Endo, Muriel Lang, Sandra Hirche, Urban Fietzek, and Dana Kulić. Data augmentation of wearable sensor data for parkinson’s disease monitoring using convolutional neural networks. In *Proceedings of the 19th ACM International Conference on Multimodal Interaction (ICMI)*, pages 216–220, 2017. doi: 10.1145/3136755.3136817.
- [20] Oresti Banos, Juan-Manuel Galvez, Miguel Damas, Hector Pomares, and Ignacio Rojas. Window size impact in human activity recognition. *Sensors*, 14(4):6474–6499, 2014. doi: 10.3390/s140406474.
- [21] Sohrab Saeb, Luca Lonini, Arun Jayaraman, David C. Mohr, and Konrad P. Kording. The need to approximate the use-case in clinical machine learning. *GigaScience*, 6(5), 2017. doi: 10.1093/gigascience/gix019.